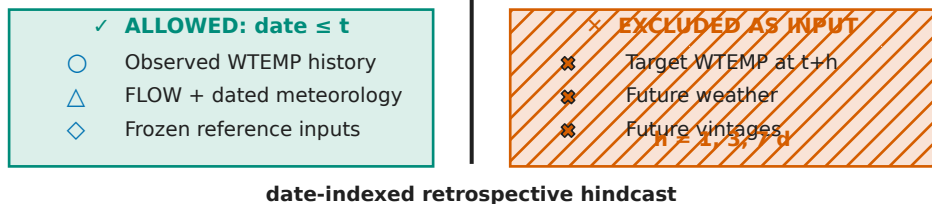


(a) Frozen role chronology



(b) Issue-time boundary



(c) Predictor source rules

VARIABLE	PROVIDER	SOURCE DATE
● WTEMP	NWIS	$\text{date} \leq t$
■ FLOW	NWIS	$\text{date} \leq t$
▲ TEMP	Daymet	$\text{date} \leq t$
◆ PRCP	Daymet	$\text{date} \leq t$
● RHMEAN	Daymet	$\text{date} \leq t$
■ DH	Daymet	$\text{date} \leq t$
▲ WDSP	gridMET	$\text{date} \leq t$

vintage: latest values served at retrieval

(d) Outcome-free predictor bridge: capability and limit



bound report + 120-request source map • latest retrospective retrieval